

Sesigo Data Portal

System Explanation and Component Map

What it is, how it works, and how it is used



Figure 1: End-to-end Sesigo Data Portal system map.

Document Control

Item	Description
Document purpose	Explain the Sesigo Data Portal in business, user, data, and technical terms.
Audience	BONASO management, system developers, coordinators, sub-grantees, funders, and system users.
Scope	Portal purpose, programme hierarchy, user roles, data workflow, live/training separation, core components, and technical architecture.
Source basis	User-provided Sesigo system map, current Sesigo branding, and previous implementation decisions shared for the portal.
Naming standard	Use Sesigo Data Portal for the platform, Sesigo Live System for production, Sesigo Training Mode for training, and Powered by BONASO as the footer/tagline.

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1. Executive Summary

The Sesigo Data Portal is a project-scoped information management and reporting platform used by BONASO to manage funded civil society organisation programme data. It brings together clients or funders, projects, BONASO oversight, coordinator organisations, sub-grantees, assigned indicators, data capture, review, dashboards, exports, and analytics in one controlled system.

The portal is not only a data-entry tool. It is a structured reporting environment that controls who can report, what they can report, which project they report under, and whether the data belongs to the official live system or the training environment.

The core design rule is simple: a client funds a project, BONASO oversees the project, coordinator organisations manage delivery, sub-grantees capture assigned indicators, and approved data feeds dashboards, reports, exports, and analytics.

A second core rule is that training data must never mix with live data. Sesigo Training Mode uses /training routes and training-only request logic. Sesigo Live System uses normal production routes. The backend remains the final authority by filtering reads and blocking cross-mode writes.

2. What the Sesigo Data Portal Is

The Sesigo Data Portal is a web-based data management system for planning, capturing, reviewing, and analysing project implementation data across BONASO-supported programmes. It provides a single place where project structures, organisation roles, reporting assignments, indicator results, narrative reports, dashboards, and exports can be managed.

From a business point of view, Sesigo is a programme coordination and reporting platform. From a technical point of view, it is a Next.js frontend connected to a Django REST API and PostgreSQL database, with controlled uploads and reporting outputs.

2.1 The portal in one sentence

Sesigo Data Portal helps BONASO and its partners manage funded project reporting by connecting projects, organisations, indicators, data capture, validation, and dashboards under one governed digital system.

3. Why the Portal Exists

The portal exists to solve fragmented data management. Without a central system, data may be captured in different Excel files, emails, paper tools, or organisation-specific templates. This creates inconsistent indicator names, duplicate records, weak version control, delayed reporting, and difficulty seeing progress across coordinators and sub-grantees.

Sesigo improves this by standardising programme structure, assigning indicators to the correct organisations, enforcing project scope, supporting review before reporting, and producing dashboards and exports from approved operational data.

3.1 Problems addressed

- Fragmented reporting across multiple organisations and templates.
- Difficulty linking data to the correct project, funder, organisation, and indicator.
- Indicator duplication caused by different wording for similar measures.
- Weak visibility of coordinator and sub-grantee performance.
- Risk of training or demo records contaminating official reporting data.
- Slow reporting caused by manual consolidation and repeated Excel clean-up.

3.2 Value delivered

- Centralised information management for BONASO and funded partners.
- Clear project and organisation accountability.
- Better data quality through validation, review, and approval workflows.
- Real-time or near real-time dashboards for management and funder reporting.
- Consistent reporting structure across live operations and training practice.

4. Who Uses the Portal

The portal is used by different people at different levels of responsibility. A single user may have a platform-level role and also participate in a project through an organisation-level role.

User group	Main purpose in the portal	Typical activities
BONASO admin / system administrator	Configure and maintain the platform.	Create users, manage roles, configure projects, assign organisations, monitor data quality, reset training data, manage system settings.
BONASO programme manager / officer	Oversee project delivery and reporting.	Review submissions, monitor indicators, follow up coordinators, view dashboards, export data.
Coordinator organisation	Coordinate delivery under a project.	Manage sub-grantees, report assigned indicators, review sub-grantee progress, submit reports.
Sub-grantee organisation	Capture project implementation data.	Enter aggregate data, capture respondents or events where applicable, upload narrative reports, view own reporting status.
Data collector / field user	Capture field-level data in allowed scope.	Submit data against assigned indicators and projects.
Client / funder viewer	View approved project performance.	Access read-only reports, dashboards, exports, and high-level progress summaries.

5. Programme and Funding Structure

The Sesigo programme model starts with a client or funder. The client funds a project. The project is overseen by BONASO. Implementation happens through coordinator organisations, which may manage sub-grantees. A coordinator can also implement some indicators directly when the project assigns them that responsibility.

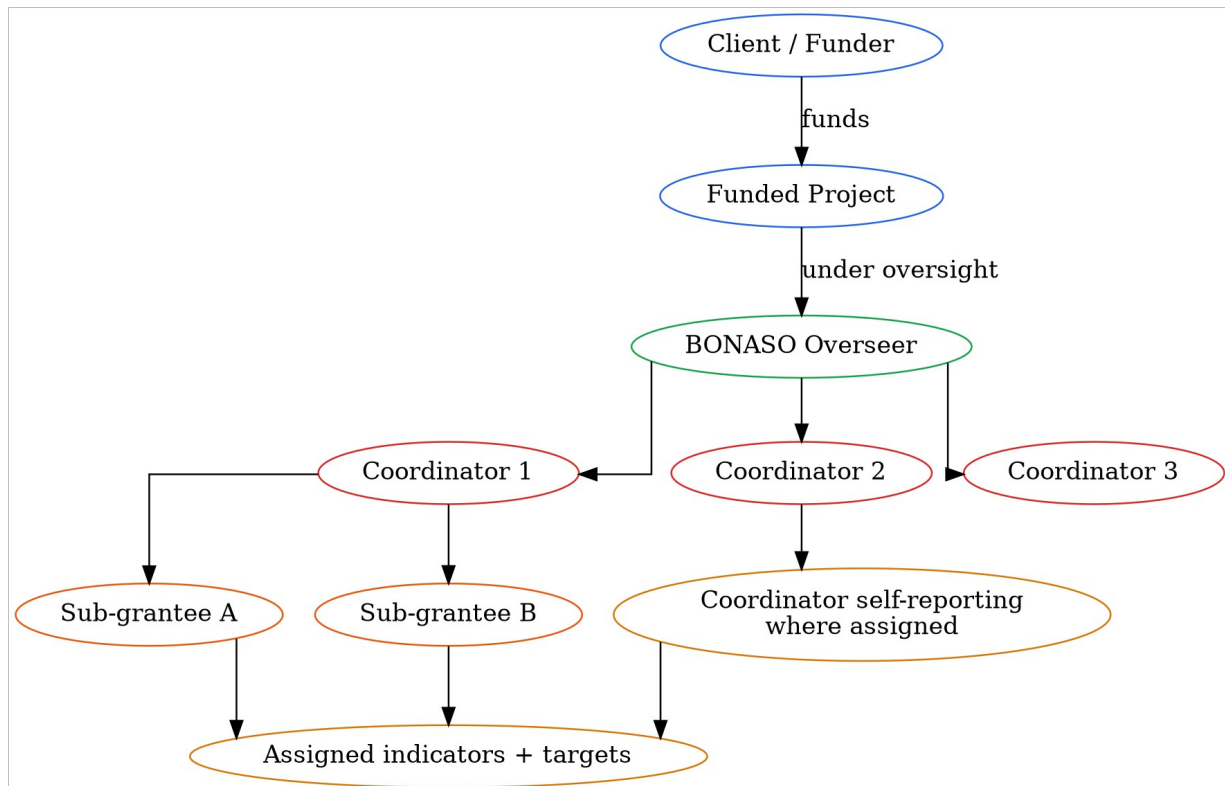


Figure 2: Client, project, BONASO, coordinator, and sub-grantee hierarchy.
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5.1 Explanation of the structure

Component	Meaning in Sesigo	Why it matters
Client / Funder	The organisation or funding source that finances the project.	Allows the portal to link project performance and reporting back to the funding source.
Project	The official reporting boundary for a funded programme.	All targets, indicators, organisations, dashboards, and training/live status are scoped through the project.
BONASO Overseer	The organisation responsible for governance, oversight, coordination, and accountability.	Ensures project delivery is monitored and reported consistently.
Coordinator organisation	A lead implementing organisation under the project.	Coordinates sub-grantees, may submit own reports, and helps manage delivery.
Sub-grantee organisation	An organisation managed by a coordinator and assigned specific indicators.	Captures service delivery data and contributes to project results.
Coordinator direct implementation	A coordinator reporting its own assigned indicators.	Supports cases where the coordinator is both manager and implementer.

6. Main System Components

The portal is made up of business components and technical components. Business components represent programme realities such as projects, organisations, indicators, targets, reports, and dashboards. Technical components ensure secure access, validation, storage, training isolation, and exports.

Component	What it does	Important rules
Clients / Funders	Stores the funder or client linked to a project.	A project should be traceable back to its funding source.
Projects	Defines the official reporting boundary, including status and training/live flag.	Every major record must be linked to the correct project.
Organisations	Stores BONASO, coordinators, sub-grantees, and implementing partners.	Users and reports are scoped through organisation relationships.
Project organisations	Defines how an organisation participates inside a specific project.	The same organisation can have different roles in different projects.
Organisation hierarchy	Represents parent-child relationships between coordinators and sub-grantees.	A coordinator should only manage/view allowed sub-grantee data.
Indicators	Defines standard performance measures.	Indicator wording must be controlled to reduce duplicates.
Project indicators	Links indicators to a project with targets and reporting expectations.	Targets should be project-specific and time-bound where needed.
Indicator assignments	Controls which organisation is allowed to report which indicator.	Organisations should not report indicators they were not assigned.
Aggregates	Stores summary data values for indicators and periods.	Records must pass validation and approval before appearing in official dashboards.
Respondents	Stores individual-level or beneficiary profile data where applicable.	Should be handled carefully with appropriate access and confidentiality.
Interactions / Responses	Stores services, contacts, responses, and event-linked data.	Must remain linked to the correct respondent, project, event, and indicator.
Events / Participants	Manages activities, sessions, attendees, and related records.	Event data must follow the same project and training/live boundary.
Narrative reports / uploads	Stores supporting reports and documents.	Uploads must be linked to project and organisation scope.
Dashboards / analytics	Turns approved data into visual performance information.	Should use approved, scoped, and correctly filtered data only.
Exports	Allows authorised users to download reporting datasets.	Exported data must respect user role, organisation scope, and mode.
Training Mode	Provides a safe environment for practice and demonstrations.	Training records must never mix with live official records.

7. How the Portal Works from Setup to Dashboard

The system follows a controlled workflow. First, the project structure is configured. Then indicators and targets are assigned. Organisations capture data only against allowed indicators. The backend validates the submission. Records become pending, are reviewed, and approved data is used for reporting outputs.

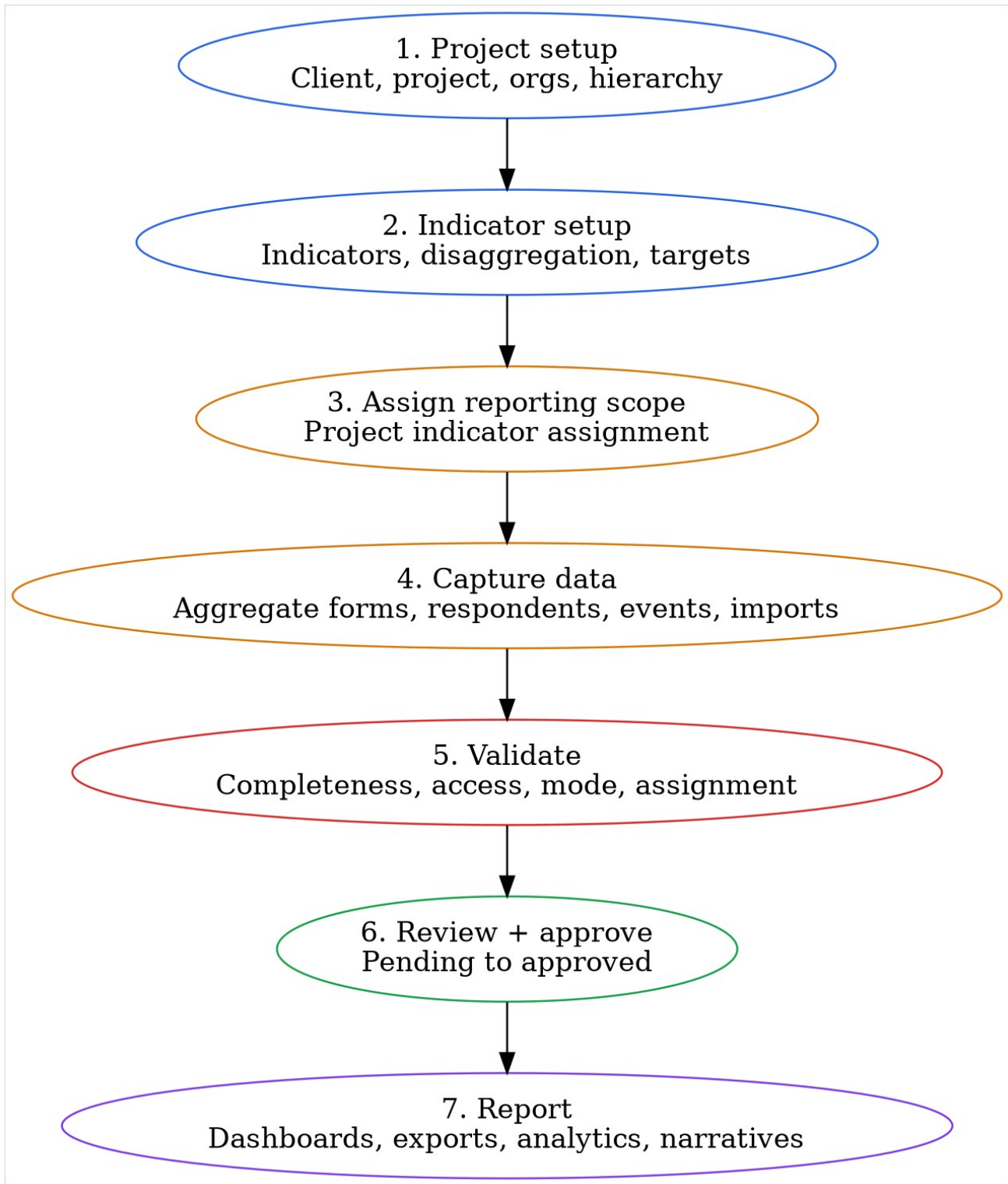


Figure 3: Data workflow from project setup to dashboards and exports.

7.1 End-to-end process

Step	Process	Description
1	Project setup	Create the funded project, define client/funder, status, dates, and whether it is live or training.
2	Organisation setup	Add BONASO oversight, coordinator organisations, sub-grantees, and their

		project-specific roles.
3	Indicator setup	Define indicators, disaggregation needs, targets, and project-specific reporting expectations.
4	Assignment	Assign indicators to organisations so each coordinator or sub-grantee knows what to report.
5	Data capture	Users enter aggregates, respondent services, event participation, workbook imports, or narrative reports.
6	Validation	System checks project scope, user scope, organisation scope, indicator assignment, and training/live mode.
7	Review and approval	Authorised users review pending records and approve valid data for official reporting.
8	Reporting	Approved records feed dashboards, reports, analytics, and exports.

8. Technical Architecture

The technical architecture separates the user interface, API logic, data storage, and reporting outputs. The frontend provides the user experience. The API enforces business rules. PostgreSQL stores operational data. Upload storage holds narrative reports and files. Dashboards and exports read from scoped, validated data.

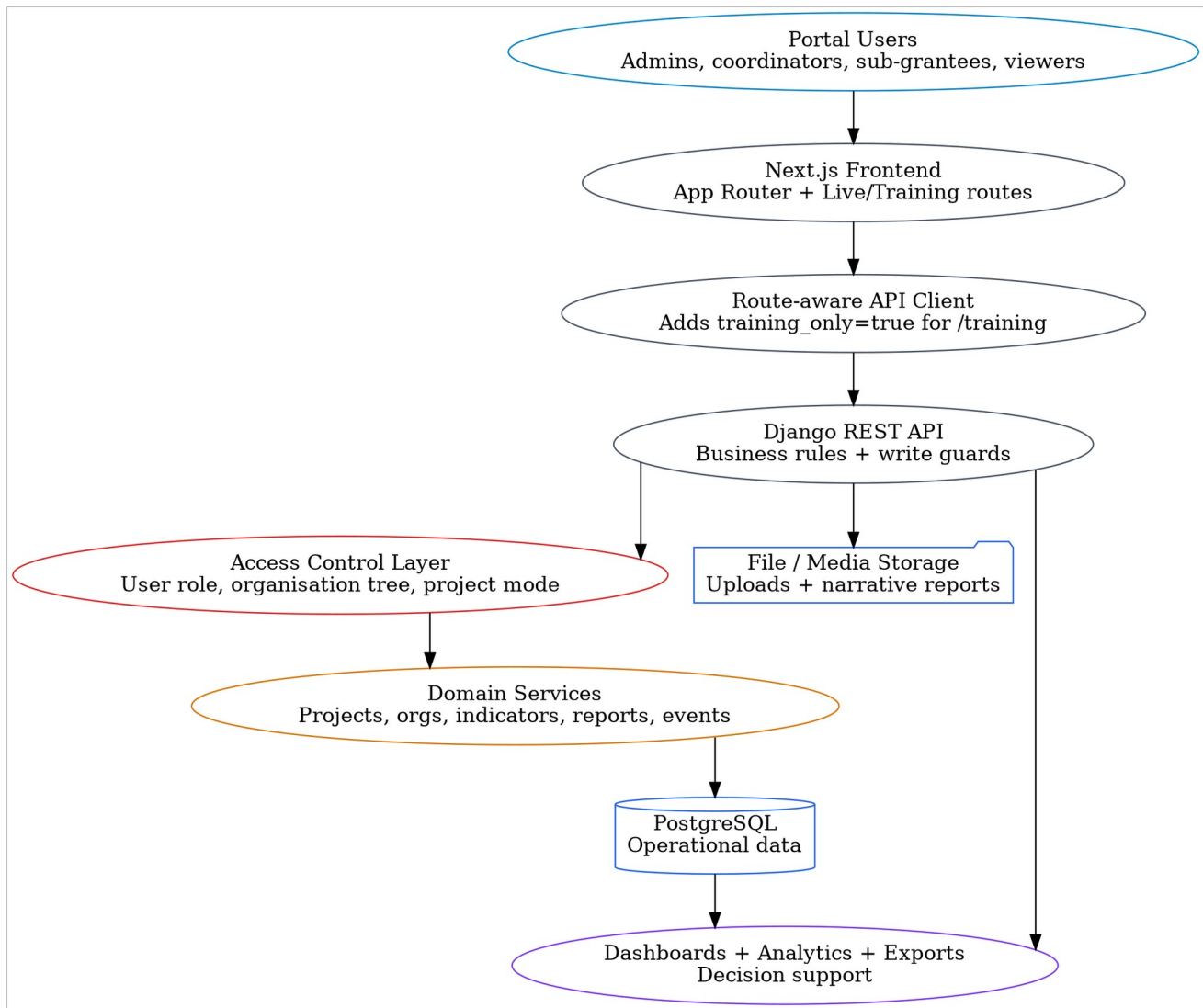


Figure 4: High-level technical architecture.

8.1 Technical component explanation

Technical layer	Description	Responsibility
Next.js frontend	The web interface users interact with.	Shows forms, dashboards, reports, live routes, and training routes.
Route-aware API client	Frontend request handler that understands whether a user is in live or training routes.	Adds training_only=true when users operate under /training routes.
Django REST API	Backend service layer that exposes data and business operations.	Validates users, roles, projects, organisation scope, indicators, and live/training boundaries.
Access control layer	Role, permission, organisation tree, project scope, and mode filtering.	Prevents users from viewing or editing data outside their allowed scope.
PostgreSQL database	Primary operational database.	Stores users, organisations, projects, indicators, targets, aggregate data, respondents, events, and reporting records.
Uploads/media storage	Storage for uploaded reports and supporting files.	Keeps narrative reports and attachments linked to projects and organisations.
Dashboards/exports/analytics	Reporting and decision-support outputs.	Uses approved, filtered, and correctly scoped data.

9. User Roles and Permissions

Sesigo uses two role concepts. The first is the platform role, such as admin, manager, officer, collector, or viewer. The second is the project-specific organisation role, such as lead, coordinator, sub-grantee, implementer, reviewer, or funder. This matters because the same organisation can play different roles in different projects.

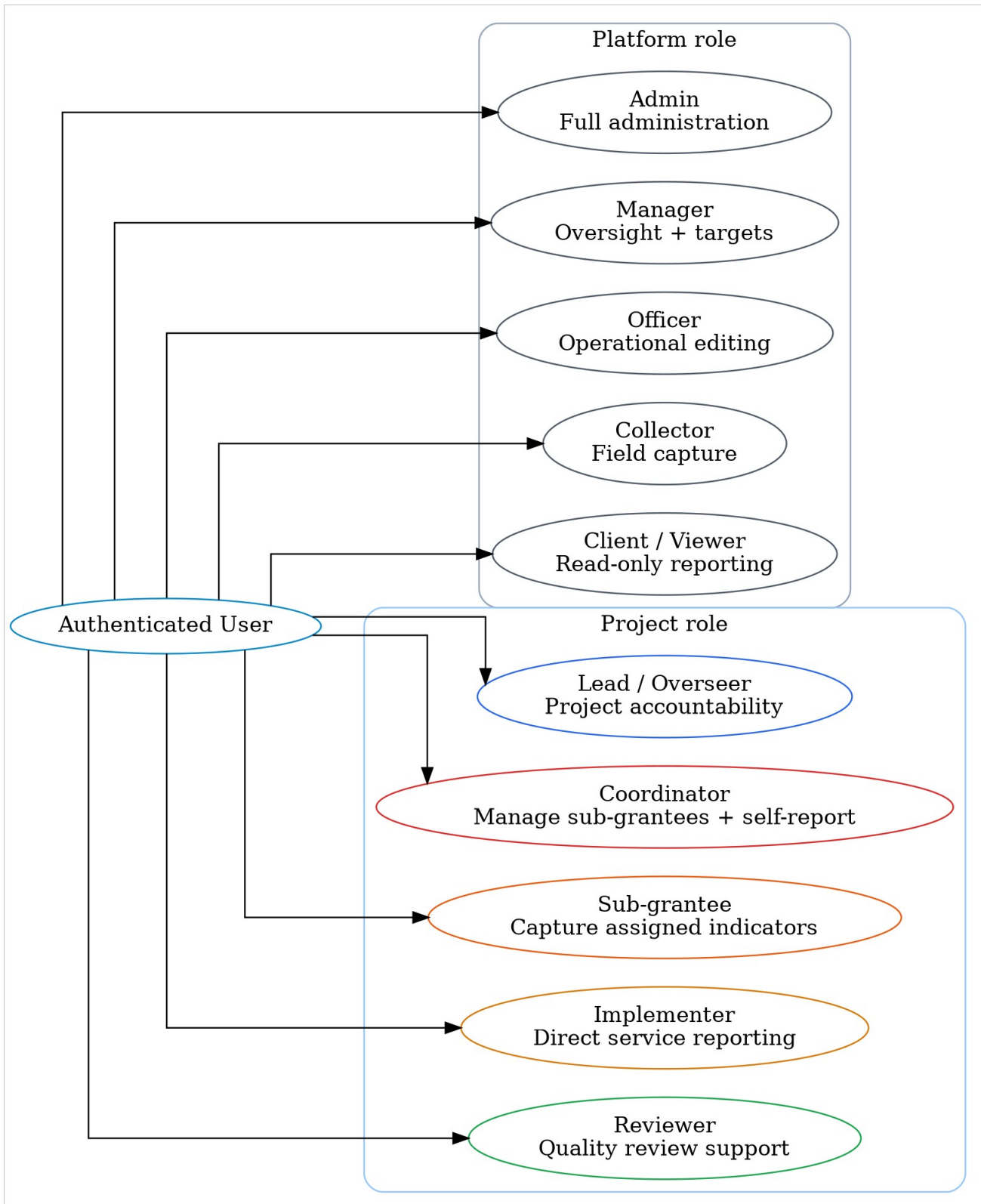


Figure 5: Platform roles and project-scoped roles.

9.1 Practical permission behaviour

- Admins can configure the system, manage users, create projects, assign organisations, review data, and manage training reset activities.
- Managers and officers can oversee reporting and operate within their assigned project and organisation scope.
- Coordinators can manage sub-grantees and may also submit their own assigned indicators.
- Sub-grantees capture data only for indicators assigned to them in the project.
- Viewers or funder users should normally access approved dashboards, reports, and exports without editing operational data.
- The backend must enforce permissions even if the frontend hides buttons or screens.

10. Data Capture, Indicators, and Disaggregation

Indicators are the centre of reporting. A project defines which indicators it tracks, sets targets, and assigns those indicators to the correct coordinator or sub-grantee. Data capture should follow the project indicator assignment rather than allowing unrestricted reporting.

10.1 Indicator structure

The recommended structure is: standard indicator -> project indicator -> organisation assignment -> period data -> review status -> dashboard output. This keeps the same indicator reusable across projects while allowing targets, assignments, and disaggregation to differ per project.

10.2 Cross-cutting indicators

Some health areas cut across target groups. For example, people living with HIV may also need NCD education and assistance. The portal should therefore avoid creating many duplicate indicators with slightly different wording. Instead, it should use canonical indicator names and capture target group, health area, age, sex, and key population as disaggregation or classification fields where appropriate.

10.3 Important programme areas and target groups

Area	Examples / notes
Health areas	HIV, NCD, mental health, human rights, substance and alcohol abuse, sexual reproductive health, and GBV.
Coordinator clusters	Male engagement and faith-based, people living with HIV, NCD, adolescents and young people, testing and counselling, and key/vulnerable populations.
Age disaggregation	Use standard age bands; adolescents and young people can be calculated from ages 10-24 where required.
Sex disaggregation	Use sex fields to calculate male engagement and other sex-specific outputs.
PLHIV disaggregation	Allow PLHIV views by age, sex, key population, and service area where the data is captured.

11. Reports, Dashboards, Analytics, and Exports

Reports and dashboards are only as strong as the data pipeline behind them. Sesigo should present approved, scoped, and correctly filtered data. Users should be able to filter results by project, organisation, indicator, reporting period, age, sex, key population, and other approved disaggregation fields.

11.1 Dashboard responsibilities

- Show project-level performance against targets.

- Show coordinator and sub-grantee contribution clearly.
- Allow filtering without mixing unrelated projects or organisations.
- Avoid double-counting where coordinator and sub-grantee reporting overlap.
- Keep training data out of live dashboards unless an authorised admin intentionally uses an include-training read-only view.

11.2 Export responsibilities

- Exports should respect the same permissions as the on-screen dashboard.
- Exports should identify project, organisation, indicator, reporting period, and approval status.
- Exports should use standard labels so data can be reused for funder reporting and programme review.

12. Sesigo Live System vs Sesigo Training Mode

Sesigo has two operating modes. The Sesigo Live System is used for official production data. Sesigo Training Mode is used for practice, onboarding, testing, and demonstrations. These modes may run on the same codebase, but the data boundary must be enforced at the backend.

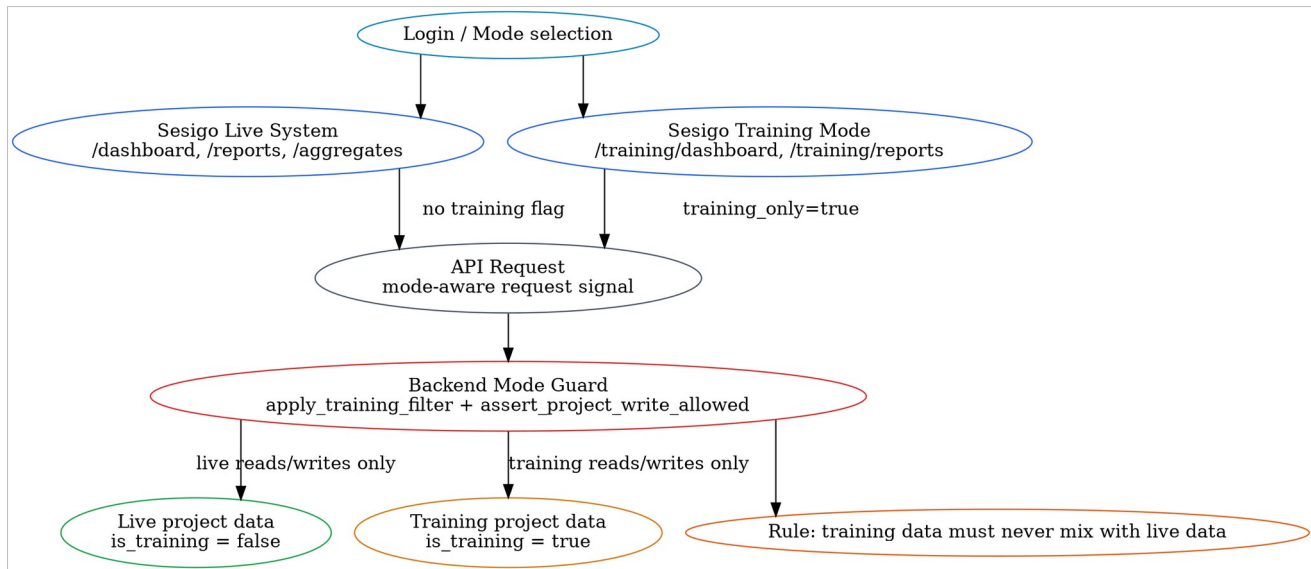


Figure 6: Live and training separation.

Concern	Sesigo Live System	Sesigo Training Mode
Route pattern	/dashboard, /reports, /aggregates, /analysis, /projects	/training/dashboard, /training/reports, /training/aggregates, /training/analysis, /training/projects
Request signal	No training flag by default.	training_only=true or mode=training request logic.
Project scope	is_training = false.	is_training = true.
Read behaviour	Excludes training-linked records by default.	Returns training-linked records only.
Write behaviour	Cannot write into training projects.	Cannot write into live projects.
Admin override	Read-only include_training=true may expose both where appropriate.	Same read-all override may apply, but writes must remain blocked across boundaries.

12.1 Why backend enforcement is important

The frontend route is helpful, but it is not enough. A user, browser error, or faulty API request can send the wrong URL or query string. For that reason the backend must always check the target project and reject cross-mode writes. This is why server-side write guards are a non-negotiable design rule.

13. Simplified Database Model

The database model is centred around the project. The project provides the reporting boundary. ProjectOrganisation defines who participates and in what role. ProjectIndicator defines what is being measured. ProjectIndicatorAssignment controls who reports which indicator. Aggregate or Interaction/Response records carry the data that feeds reports.

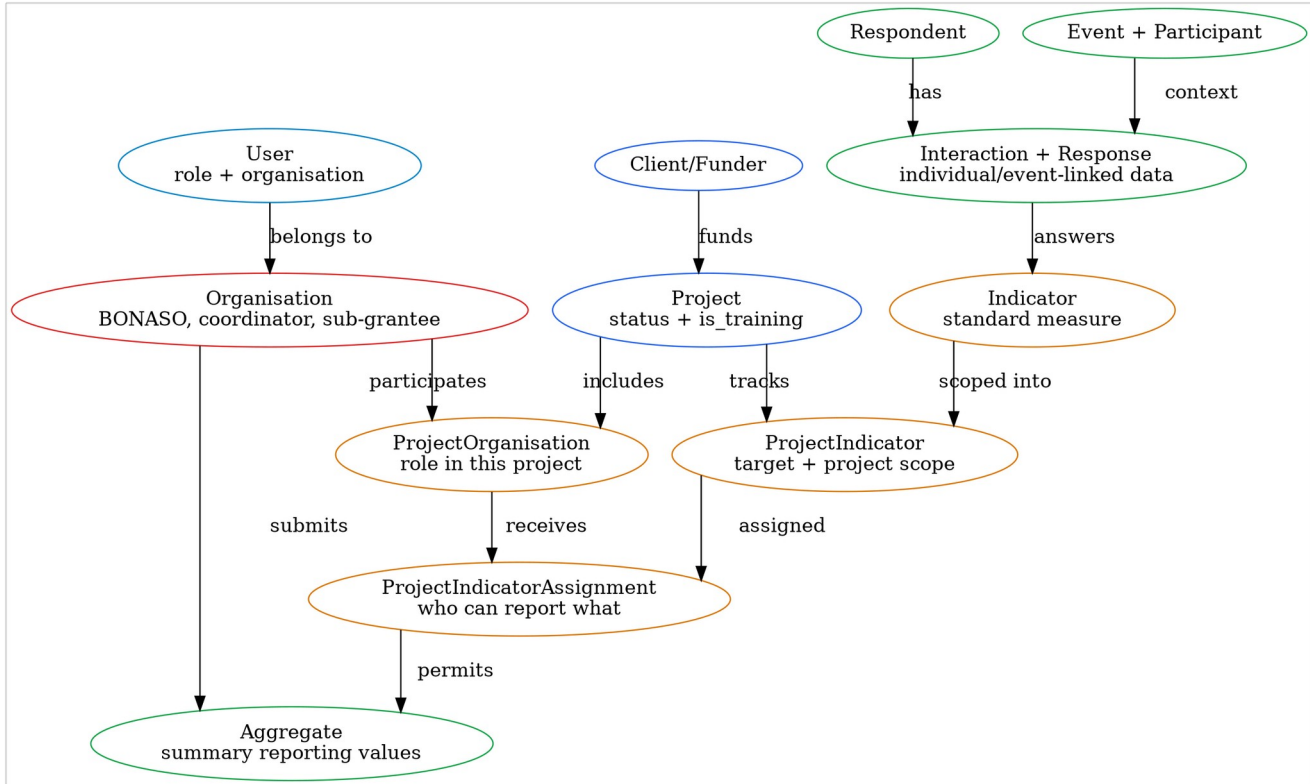


Figure 7: Simplified database model for the core reporting path.

Entity	Purpose
User	Stores the user account, platform role, and organisation link.
Organisation	Stores BONASO, coordinators, sub-grantees, and implementing partners.
Client / Funder	Stores the funding source linked to a project.
Project	Stores the funded project, including live/training status.
ProjectOrganisation	Defines an organisation role inside a specific project.
Indicator	Stores the standard indicator definition.
ProjectIndicator	Links an indicator to a project and its targets.
ProjectIndicatorAssignment	Controls which organisation can report which project indicator.
Aggregate	Stores summarised indicator results by period, project, organisation, and status.
Respondent / Interaction / Response	Stores individual-level or service-event data where applicable.
Event / Participant	Stores activity/event participation data.

14. How the System Is Used by Each User Group

14.1 System administrator

- Create and manage users.
- Create clients/funders and projects.
- Set whether a project is live or training.
- Assign organisations to projects and define hierarchy.
- Configure indicators, targets, and organisation assignments.
- Monitor logs, access, backups, training reset, and security settings.

14.2 BONASO programme officer or manager

- Review project performance across coordinators and sub-grantees.
- Monitor reporting completeness and quality.
- Approve or query submitted data.
- Use dashboards to identify gaps, trends, and under-performing areas.
- Export approved data for donor and management reporting.

14.3 Coordinator organisation

- View assigned project and sub-grantee structure.
- Submit coordinator-level data where the coordinator implements directly.
- Monitor sub-grantee submissions and follow up missing reports.
- Use dashboards to see own contribution and sub-grantee performance.

14.4 Sub-grantee organisation

- Log in to the assigned project scope.
- Capture data against assigned indicators.
- Upload supporting narrative reports where required.
- Review submission status and correct rejected or queried records.

14.5 Trainer or trainee

- Use Sesigo Training Mode for safe practice.
- Enter demo records without affecting official live data.
- Practice dashboards, reports, and exports using training records only.
- Training data can be reset or purged without affecting production data.

15. Administration, Security, and Quality Assurance

15.1 Administration responsibilities

- Keep user accounts and roles accurate.
- Keep project and organisation structures aligned with signed implementation arrangements.
- Maintain indicator catalogues and prevent duplicate indicator wording.
- Back up the database and media files regularly.
- Document deployments, configuration changes, and migrations.

15.2 Security responsibilities

- Use secure production settings for HTTPS, cookies, CSRF, and secret keys.
- Enforce backend permissions for every read and write operation.
- Keep training/live isolation server-side, not only through the UI.

- Apply least-privilege access for funders, coordinators, sub-grantees, and data collectors.
- Protect respondent-level data and avoid unnecessary exposure in exports.

15.3 Quality assurance responsibilities

- Test dashboard totals against known datasets.
- Test organisation scope and project-specific indicator assignments.
- Test training mode isolation for reads, writes, dashboards, exports, respondents, and events.
- Test import templates and validation messages before rollout.
- Use audit logs to trace who created, edited, approved, or exported records.

16. Implementation Checklist for Developers

Area	Checklist item
Project model	Every core reporting record must link to a project or be filterable through a project-linked entity.
Training/live isolation	Every read endpoint must apply training filtering; every write endpoint must call the project write guard.
Organisation scope	Non-admin users must only see organisations within their allowed project tree.
Indicator assignment	Data capture forms must check that the organisation has the indicator assignment.
Dashboards	Dashboard queries must avoid mixing training/live data and must avoid double-counting.
Exports	Exports must apply the same filters as the UI and API.
Uploads	Uploaded reports must be linked to project, organisation, user, and mode.
Audit logs	Critical actions should record user, timestamp, old value, new value, and affected record.
Testing	Write tests for permission boundaries, mode separation, and dashboard totals.
Documentation	Keep system maps, ERDs, and user guides updated when schema or workflows change.

17. Future Improvements

- Add a physical separate training database option if BONASO wants stronger infrastructure isolation in the future.
- Add full audit logs for create, update, delete, approval, export, and training reset actions.
- Strengthen project-specific indicator assignment so every reporting form is assignment-driven.
- Add better dashboard filtering by project, coordinator, sub-grantee, indicator, age, sex, key population, and health area.
- Add stronger indicator aliasing and canonical labels to reduce duplicate indicator wording.
- Add configurable reporting periods, quarterly targets, and target-versus-achievement views.
- Add training data expiry or reset rules, for example automatic purge after a defined number of days.
- Add user-facing help text and tooltips for coordinators and sub-grantees.
- Add automated QA checks for data completeness, outliers, duplicate submissions, and invalid disaggregation.

18. Architectural Conclusion

Sesigo Data Portal should be understood as a project-scoped reporting platform. Its strength comes from explicit modelling of the funder, project, BONASO oversight, coordinator organisations, sub-grantees, indicators, assignments, validation, approval, and reporting outputs.

The long-term technical direction should keep three principles at the centre: project scope must be explicit, indicator assignment must control reporting, and training data must remain isolated from live official reporting data.

When new features are added, developers should always ask: Which project does this belong to? Which organisation is allowed to see or edit it? Which indicator assignment permits the record? Is this live data or training data? If those questions are answered consistently, the Sesigo Data Portal will remain scalable, auditable, and safe for BONASO programme reporting.